through the black hole—inward near the poles and outward near the equator—seemed very peculiar. During 1977 and 1978, Znajek and, independently, Thibault Damour (a recent PhD at the Obser-vatoire de Paris in France) puzzled over this peculiarity. While trying to understand it, they independently translated the curved spacetime equations, which describe the spinning hole and its plasma and magnetic field, into an unfamiliar form with an intriguing pictorial interpretation: The current, when it reaches the horizon, does not enter the hole. Instead, it attaches itself to the horizon, where it is carried by the kinds of horizon charges previously imagined by Hanni and Ruffini. This horizon current flows from the pole to the equator, where it exits up the magnetic field lines. Moreover, Damour and Znajek discovered, the laws that govern the horizon's charge and current are elegant versions of the flat spacetime laws of electricity and magnetism: They are Gauss's law, Ampère's law, Ohm's law, and the law of charge conservation (Figure 11.5).

Damour and Znajek did not assert that a being who falls into the black hole will encounter a membrane-like horizon with electric charges and currents. Rather, they asserted that if one wishes to figure out how electricity, magnetism, and plasmas behave outside a black hole, it is useful to regard the horizon as a membrane with charges and currents.

When I read the technical articles by Damour and Znajek, I suddenly understood: They, and Hanni and Ruffini before them, were discovering the foundations of a new paradigm for black holes. The paradigm was fascinating. It captivated me. Unable to resist its allure, I spent much of the 1980s, together with Richard Price, Douglas Mac-donald, Ian Redmount, Wai-Mo Suen, Ronald Crowley, and others, bringing it into a polished form and writing a book on it, *Black Holes: The Membrane Paradigm*

The laws of black-hole physics, written in this membrane paradigm, are completely equivalent to the corresponding laws of the curved spacetime paradigm—so long as one restricts attention to the hole's exterior. Consequently, the two paradigms give precisely the same predictions for the outcomes of all experiments or observations that anyone might make outside a black hole—including all astronomical observations made from Earth. When thinking about astronomy and astrophysics, I find it useful to keep both paradigms at hand, membrane and curved spacetime, and to do Escher-type mind-flips back and forth between them. The curved spacetime paradigm, with its horizons made from curved empty spacetime, may be